Chapter 2
Asymptotic Equipartition Property and Entropy Rate（渐进等分性质和熵率）

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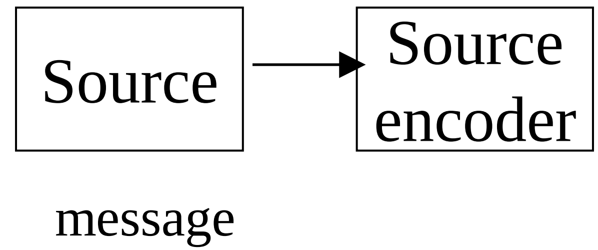
Preview

- In Chapter 1,we introduced many basic notions
  - Entropy
  - Joint Entropy and Conditional Entropy
  - Mutual Information
- These notions are used to describe some features of a random variable, or relationship of several random variables.
  随机变量或多个随机变量之间的关系
- But how to apply these basic notions to communications? (Note that we have mentioned that we are considering the information theory in communications) 如何在通信中应用？
- In this chapter, we use a discrete random variable *X* with probability mass function (pmf) *p(X)* to describe a ***source***. The outputs of the source is the realizations of the discrete random variable *X*. 用随机变量来描述信源

Preview

- Source examples
  - Character
    - ASCII: A~Z, 0~9, … 128
    - GB2312-800: 6763,
    - GBK: GB2312+BIG5, more than 30000
  - Sound
    - Voice: PCM,
    - Music: MP3.
  - Image/Video
    - Picture: BMP, JPG,
    - Movie: MPEG1(VCD), MPEG2(DVD), RMVB



- Let us consider how to describe the source by information bits more efficiently. We first consider the independent, identically distributed (i.i.d.) random variables case.（独立同分布）

Question

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- Suppose that we have a source generates independent binary information bits sequence randomly, 0100111010110…
  - If 0 → 0.55, 1 → 0.45, then as the length of the sequence goes very large, what kind of sequence we are most likely to see ?

    010011011001110001111011001100110101110010101 ?

    111111111010111110011111010111010101111111111111 ?
  - If 0 → 0.2, 1→ 0.8,   ?
  - If 0 → 0.1, 1→ 0.9

    010011011001110001111011001100110101110010101 ?

    111111111010111110011111010111010101111111111111 ?

How many bits we actually need to describe the sequences generated by such a source (or the random variable *X*) ??

Outline

- Convergence of Random Variables
- Laws of Large Numbers
- Asymptotic Equipartition Property (AEP)
- Properties
- Entropy Rates of a Stochastic Process
- AEP for Stationary Stochastic Processes

Asymptotic Equipartition Property

Laws of Large Numbers

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- ***Weak*** Law of Large Numbers（弱大数定理）

Let $X_1, X_2, \cdots X_n$ be a sequence of independent and identically distributed random variables, each having finite mean $E[X_i] = \mu$
Then, for any $\varepsilon > 0$

$$P\left(\left|\frac{X_1 + X_2 + \cdots X_n}{n} - \mu\right| \geq \varepsilon\right) \xrightarrow{n\to\infty} 0$$

- ***Strong*** Law of Large Numbers （强大数定理）

Let $X_1, X_2, \cdots X_n$ be a sequence of independent and identically distributed random variables, each having finite mean $E[X_i] = \mu$
Then,

$$P\left(\left|\lim_{n\to\infty} \frac{X_1 + X_2 + \cdots X_n}{n} - \mu\right| = 0\right) = 1$$

Converge ***in probability***
（依概率收敛）

→

Weak Law of Large Numbers
（弱大数定理）

Converge ***almost surely/ with probability 1***
（几乎处处收敛）

→

Strong Law of Large Numbers
（强大数定理）

Laws of Large Numbers

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- Differences between the weak law and the strong law
  - The *weak law* states that for a specified large *n*, the average $\overline{X}_n$ is likely to be near *μ*. Thus, it leaves open the possibility that $|\overline{X}_n - \mu| > \varepsilon$ happens an infinite number of times, although at infrequent intervals.
  - 概率趋向于0，但是实际发生的次数仍然可能是无穷大
  - The *strong law* shows that this *almost surely* will not occur. In particular, it implies that with probability 1, we have that for any *ε* > 0 the inequality $|\overline{X}_n - \mu| < \varepsilon$ holds for all large enough *n*.

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Asymptotic Equipartition Property (AEP)

- Intuitive idea of AEP
  - Assume X a R.V. so that $P(X=1)=p=1-P(X=0)$ We observe a sequence of R. V. $X_1, X_2 \cdots X_n$ with $P(X_i)=P(X)$

$$P(X_1, X_2 \cdots X_n = x_1, x_2 \cdots x_n) = p^{\sum x_i}(1-p)^{n-\sum x_i}$$

  - We can see these sequences are not equally likely 不等概, some sequences have high probability to appear. However, as $n \to \infty$, only $\approx 2^{nH(X)}$ sequences appear, and all of them appear with the same probability $\approx 1/2^{nH(X)}$
  - The typical set 典型集 is the set of all these $\approx 2^{nH(X)}$ sequences. Each called a typical sequence典型序列.
  - Since these are the only sequences that appear, we do not care about the other ones.→ we just need $nH(X)$ bits to represent all original sequences instead of n bits,

AEP

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- AEP
  - AEP:

Let $X_1, X_2 \cdots X_n$ be a sequence of i.i.d random variables with distribution $p(X)$

$$\Rightarrow -\frac{1}{n}\log p\left(X_1, X_2 \cdots, X_n\right) \xrightarrow{n\to\infty} H(X), \text{ in probability}$$

  - Proof: using weak Law of Large Numbers

Let $X_1, X_2, \cdots X_n$ be a sequence of i. i. d random variables with distribution $p(X)$

$$\frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{n\to\infty} E[X], \text{ in probability}$$

Corollary: $$\frac{1}{n}\sum_{i=1}^{n} f(X_i) \xrightarrow{n\to\infty} E[f(X)], \text{ in probability}$$

AEP

- Proof:

$$-\frac{1}{n}\log p(X_1, X_2, \ldots, X_n) = -\frac{1}{n}\sum_i \log p(X_i)$$

$$\to -E\log p(X) \qquad \text{in probability}$$

$$= H(X),$$

Corollary

$$\frac{1}{n}\sum_{i=1}^{n} f(X_i) \xrightarrow{n\to\infty} E\left[f(X)\right], \text{ in probability}$$

“AEP即为大数定理”

AEP

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- Typical set

Definition The *typical set* $A_\epsilon^{(n)}$ with respect to $p(x)$ is the set of sequences $(x_1, x_2, \ldots, x_n) \in \mathcal{X}^n$ with the property

$$2^{-n(H(X)+\epsilon)} \le p(x_1, x_2, \ldots, x_n) \le 2^{-n(H(X)-\epsilon)}.$$

或者 $A_\varepsilon^{(n)} = \left\{(x_1, x_2, \cdots, x_n): 2^{-n(H(X)+\varepsilon)} < p\left(x_1, x_2 \cdots, x_n\right) < 2^{-n(H(X)-\varepsilon)}\right\}$

或者 $A_\varepsilon^{(n)} = \left\{(x_1, x_2, \cdots, x_n): \left|-\frac{1}{n}\log p\left(x_1, x_2 \cdots, x_n\right) - H(X)\right| < \varepsilon\right\}$

- Interpretation
  - Observe sequence $\left(x_1, x_2 \cdots x_n\right)$, if $n \to \infty$, we will see $\left(x_1, x_2 \cdots x_n\right) \in A_\varepsilon^{(n)}, \forall \varepsilon > 0$ and $p\left(x_1, x_2 \cdots x_n\right) \simeq 2^{-nH(X)}, as\ \varepsilon \to 0$

    We will see $\left|A_\varepsilon^{(n)}\right| \simeq 2^{nH(X)}$

    随着观察序列的增长，我们只能看到典型序列出现，并且各自的概率接近等概。

AEP

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- Properties of typical set 典型集合的性质**1-4**

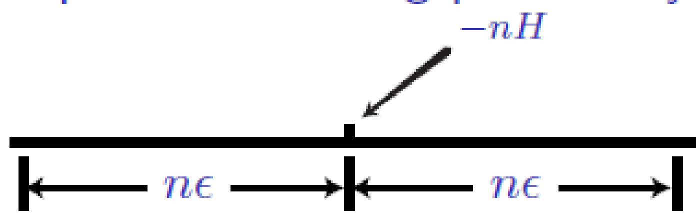
1. If $(x_1, x_2, \ldots, x_n) \in A_\epsilon^{(n)}$, *then*

$$H(X) - \epsilon \le -\frac{1}{n}\log p(x_1, x_2, \ldots, x_n) \le H(X) + \epsilon.$$

Proof: It is immediate from the definition of typical set $A_\varepsilon^{(n)}$

$$A_\varepsilon^{(n)} = \left\{ (x_1, x_2, \cdots, x_n) : 2^{-n(H(X)+\varepsilon)} < p\left(x_1, x_2 \cdots, x_n\right) < 2^{-n(H(X)-\varepsilon)} \right\}$$

Typical set are those sequences with log probability within the range



AEP

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2. $\Pr\left\{A_\epsilon^{(n)}\right\} > 1-\epsilon$ *for n sufficiently large.*

Proof: from the definition of AEP $-\frac{1}{n}\log p\left(x_1, x_2 \cdots, x_n\right) \xrightarrow{p} H(X)$

Definition： $\forall \varepsilon > 0, \delta > 0, \exists n_0 = N, \forall n > n_0$

$$P\left(\left|-\frac{1}{n}\log p\left(X_1, X_2 \cdots, X_n\right) - H(X)\right| < \varepsilon\right) > 1-\delta$$

$$\Rightarrow P\left(A_\varepsilon^{(n)}\right) > 1-\varepsilon, \quad \text{if take } \delta = \varepsilon$$

The typical set has probability nearly 1　　只有典型序列才会出现

$$\Pr\left\{(X_1, X_2, \ldots, X_n) : p(X_1, X_2, \ldots, X_n) = 2^{-n(H\pm\epsilon)}\right\} \approx 1$$

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3. $\left|A_\epsilon^{(n)}\right| \le 2^{n(H(X)+\epsilon)}$, *where* $|A|$ *denotes the number of elements in the set* A.

$$1 = \sum_{\mathbf{x}\in\mathcal{X}^n} p(\mathbf{x}) \ge \sum_{\mathbf{x}\in A_\epsilon^{(n)}} p(\mathbf{x}) \ge \sum_{\mathbf{x}\in A_\epsilon^{(n)}} 2^{-n(H(X)+\epsilon)}$$

$$= 2^{-n(H(X)+\epsilon)}|A_\epsilon^{(n)}|,$$

The number of elements in the typical set is a bit less than $2^{nH(X)}$

典型序列总数稍小于 $2^{nH(X)}$

AEP

4. $|A_\epsilon^{(n)}| \geq (1-\epsilon)2^{n(H(X)-\epsilon)}$ *for n sufficiently large.*

$$1-\epsilon < \Pr\{A_\epsilon^{(n)}\} \leq \sum_{\mathbf{x}\in A_\epsilon^{(n)}} 2^{-n(H(X)-\epsilon)} = 2^{-n(H(X)-\epsilon)}|A_\epsilon^{(n)}|$$

$$\sum_{\mathbf{x}\in A_\epsilon^{(n)}} p(\mathbf{x})$$

典型序列总数稍大于 $2^{nH(X)}$

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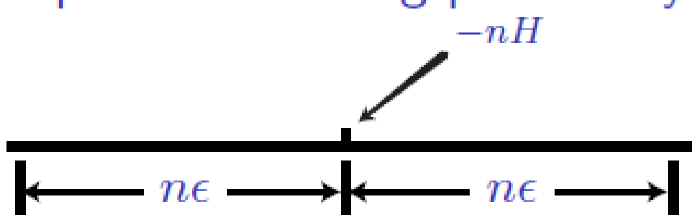
- From property 3 and 4, if n sufficiently large, $\varepsilon \to 0$

$$\left|A_\varepsilon^{(n)}\right| \simeq 2^{nH(X)}$$

- From property 2, if n sufficiently large and I observe sequence $\left(x_1, x_2 \cdots x_n\right) \in A_\varepsilon^{(n)}$ with probability 1.
- From property 1, All sequences in typical set have the same probability $\simeq 2^{-nH(X)}$

Typical set are those sequences with log probability within the range



AEP

- Assume X a R.V. so that $P(X=1)=\boxed{p<\frac{1}{2}}$

$$H(X)=-p\log p-(1-p)\log(1-p)$$

- The typical set $A_{\epsilon}^{(n)}$ is the set of sequence with about pn ones and $(1-p)n$ zeros.
- The probability of a typical sequence is about $p^{pn}(1-p)^{(1-p)n}=2^{-nH(X)}$
- The number of sequence with pn ones is $C_n^{pn}=\dfrac{n!}{(pn)!(n-pn)!}$
- There are totally 2^n sequences, but

  most of them are collectively very improbable.

Example $P(X=1)=0.1,\ P(X=0)=0.9 \qquad H(X)\approx 0.469$

$$n=100 \qquad 2^{100}\approx 10^{30} \qquad 2^{nH}=10^{14}\ll 10^{30}$$

So the number of typical sequences is much smaller than the number of possible sequences.典型序列的数量远小于序列总数

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- Note that the most probable sequence have almost all zeros, but they are not in the typical set. ---- there are not enough of them to matter.（概率最大的那个序列是全零序列，但是其不在典型集中）

Most likely sequence has probability $0.9^{100} \approx 2.66 \times 10^{-5}$

but a typical sequence has probability 7.62×10^{-15}

There are **exponentially many sequences** that are typical than there are “high" probability sequences.

The probability of each individual sequence goes to zero, but the size of the set of typical sequences grows fast enough $n \to \infty$

AEP

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- High Probability Sets （高概率集合）
  - Typical set is a fairly small set that contains most of the probability.
  - It is not clear whether it is the smallest such set.
  - Smallest set

Definition For each $n = 1, 2, \ldots,$ let $B_\delta^{(n)} \subset \mathcal{X}^n$ be the smallest set with

$$\Pr\{B_\delta^{(n)}\} \ge 1 - \delta.$$

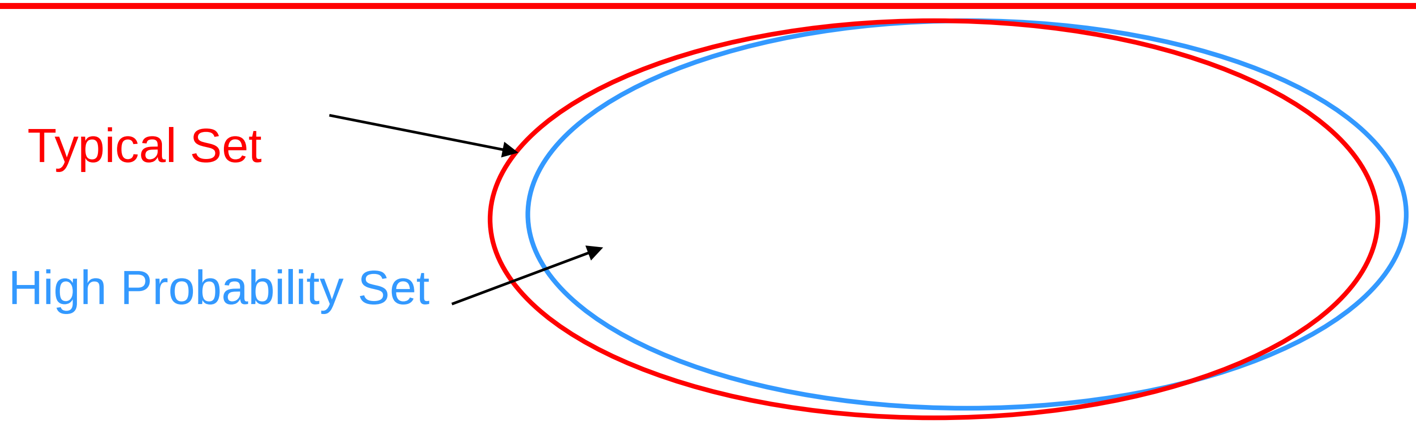
  - The smallest set is different from the typical set, but must have significant intersection with and therefore must have about as many elements.（两者定义不同，但是含有的元素个数基本一样，且存在显著交集）

AEP

Theorem 3.3.1 *Let* $X_1, X_2, \ldots, X_n$ *be i.i.d.* $\sim p(x)$. *For* $\delta < \frac{1}{2}$ *and any* $\delta' > 0$, *if* $\Pr\{B_\delta^{(n)}\} > 1 - \delta$, *then*

$$\frac{1}{n} \log |B_\delta^{(n)}| > H - \delta' \qquad \textit{for } n \textit{ sufficiently large.}$$

$B_\delta^{(n)}$ must have at least 2^{nH} elements, is about the same size as the $A_\epsilon^{(n)}$



AEP

- The AEP establishes that $nH(X)$ bits suffice on the average to describe n **independent identically distributed** random variables. （前面讨论的是独立同分布随机变量的情况）

- But what if the random variables are dependent? In particular, what if the random variables form a stationary process?

In the following, we will show, just as in the i. i. d. case, that the entropy $H(X_1, X_2, \cdots, X_n)$ grows (asymptotically) linearly with n at a rate $H(\mathcal{X})$, which we will call the *entropy rate of the stochastic (random) process*. *（随机过程的熵率）*

Entropy Rate

Stochastic (Random) Process

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- Stochastic process（随机过程）
  - Definition: a stochastic process is just a time sequence of random variables.
  - $\{X_i\}_{i\in I}$ or $X(t)$, at each *n* or *t*, is a random variable.
  - To describe a stochastic process, we use the ***joint distribution***

$$P\left(X_1 = x_1, X_2 = x_2 \cdots X_n = x_n\right)$$

- Stationary（平稳）
  - A stochastic process is stationary when the joint probability mass of any subset of random variables is invariant with respect to shifts in the time index. (严平稳定义）

$$P\left(X_1 = x_1, X_2 = x_2 \cdots X_n = x_n\right) = P\left(X_{1+i} = x_1, X_{2+i} = x_2 \cdots X_{n+i} = x_n\right)$$

  - The stationary property makes a stochastic process easy to analyze → time index is not needed.

Stochastic Process

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- Ergodic Property（各态历经）
  - A stationary process is ergodic if it satisfies the law of large numbers

Mean：
均值

$$\frac{1}{n}\sum_{i=1}^{n} X_i \xrightarrow{n\to\infty} E[X], \text{ with probability } 1$$

Time average

Ensemble average (calculated in prob. space)

时间平均

集平均

Auto-correlation Function
自相关函数

$$\frac{1}{n}\sum_{i=1}^{n} X_i X_{i+k} \xrightarrow{n\to\infty} E[X_i X_{i+k}], \text{ with probability } 1$$

Time average

Ensemble average (calculated in prob. space)

Entropy Rates of a Stochastic Process

- Entropy Rate
  - Measure of how $H(X_1, X_2 \cdots, X_n)$ **grow** with n. (Rate of growth增长率)

Definition The *entropy* of a stochastic process $\{X_i\}$ is defined by

$$H(\mathcal{X}) = \lim_{n\to\infty} \frac{1}{n} H(X_1, X_2, \ldots, X_n)$$

when the limit exists.

  - Simple examples
    - i. i. d R. V. $X_1, X_2 \cdots, X_n$

$$H(\mathcal{X}) = \lim \frac{H(X_1, X_2, \ldots, X_n)}{n} = \lim \frac{nH(X_1)}{n} = H(X_1)$$

    - Independent but not identically distributed $X_1, X_2 \cdots, X_n$

$$H(X_1, X_2, \ldots, X_n) = \sum_{i=1}^{n} H(X_i), \quad \lim_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n} H(X_i)$$

Entropy Rates of a Stochastic Process

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- identically distributed but not independent $X_1, X_2 \cdots, X_n$

$$H(\mathcal{X}) = \lim \frac{H(X_1, X_2, \ldots, X_n)}{n} = \lim \frac{1}{n} \sum_{i=1}^{n} H\left(X_i \middle| X_{i-1} \cdots, X_1\right)$$

$$\leq \lim \frac{1}{n} \sum_{i=1}^{n} H\left(X_i\right) = \lim \frac{1}{n} \sum_{i=1}^{n} H\left(X\right) = H\left(X\right)$$

- Another definition

$$H'(\mathcal{X}) = \lim_{n \to \infty} H(X_n | X_{n-1}, X_{n-2}, \ldots, X_1)$$ when the limit exists.

Interpretation: Growing of uncertainty when add a new variable.

The first is the per symbol entropy of the *n* random variables,
The second is the conditional entropy of the last random variable given the past.

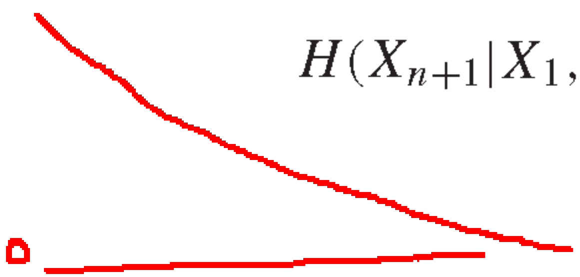
Entropy Rates of a Stochastic Process

For a stationary stochastic process, the two limits exist and are equal.

$$H(\mathcal{X}) = H'(\mathcal{X}).$$

- Proof: two steps
  - Limit exists

Conditioning reduces entropy



$$H(X_{n+1}|X_1, X_2, \ldots, X_n) \le H(X_{n+1}|X_n, \ldots, X_2)$$
$$= H(X_n|X_{n-1}, \ldots, X_1),$$

Stationarity of the process

A decreasing sequence of nonnegative numbers, the limit always exists 一个递减的非负数列，极限总是存在

  - Equality

Entropy Rates of a Stochastic Process

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Theorem 4.2.3 *(Cesáro mean)* *If* $a_n \to a$ *and* $b_n = \frac{1}{n}\sum_{i=1}^{n} a_i$, *then* $b_n \to a$.

Formal Proof: Let $\epsilon > 0$. Since $a_n \to a$, there exists a number $N(\epsilon)$ such that $|a_n - a| \le \epsilon$ for all $n \ge N(\epsilon)$. Hence,

$$
\begin{aligned}
|b_n - a| &= \left| \frac{1}{n} \sum_{i=1}^{n} (a_i - a) \right| \\
&\le \frac{1}{n} \sum_{i=1}^{n} |(a_i - a)| \\
&\le \frac{1}{n} \sum_{i=1}^{N(\epsilon)} |a_i - a| + \frac{n - N(\epsilon)}{n}\epsilon \\
&\le \frac{1}{n} \sum_{i=1}^{N(\epsilon)} |a_i - a| + \epsilon
\end{aligned}
$$

$$H'(\mathcal{X}) = \lim_{n\to\infty} H(X_n|X_{n-1}, X_{n-2}, \ldots, X_1)$$

$$
\begin{aligned}
H(\mathcal{X}) &= \lim \frac{H(X_1, X_2, \ldots, X_n)}{n} \\
&= \lim \frac{1}{n}\sum_{i=1}^{n} H\left(X_i \middle| X_{i-1} \cdots, X_1\right)
\end{aligned}
$$

Entropy Rates of a Stochastic Process

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- Basic idea: Since most of the terms in the sequence $\{a_k\}$ are eventually close to a, then the average of the first n terms is also eventually close to a

By the chain rule,
$$\frac{H(X_1, X_2, \ldots, X_n)}{n} = \frac{1}{n}\sum_{i=1}^{n} H(X_i|X_{i-1}, \ldots, X_1),$$

The entropy rate is the time average of the conditional entropies. The conditional entropies tend to a limit H'.

$$\begin{aligned} H(\mathcal{X}) &= \lim \frac{H(X_1, X_2, \ldots, X_n)}{n} = \lim H(X_n|X_{n-1}, \ldots, X_1) \\ &= H'(\mathcal{X}). \qquad \square \end{aligned}$$

Entropy Rates of a Stochastic Process

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1	space	0.1817	15	M	0.02105
2	E	0.1073	16	U	0.02010
3	T	0.0856	17	G	0.01633
4	A	0.0668	18	Y	0.01623
5	O	0.0654	19	P	0.01623
6	N	0.0581	20	W	0.01260
7	R	0.0559	21	B	0.01179
8	I	0.0510	22	V	0.00752
9	S	0.0499	23	K	0.00344
10	H	0.04305	24	X	0.00136
11	D	0.03100	25	J	0.00108
12	L	0.02775	26	Q	0.00099
13	F	0.02395	27	Z	0.00063
14	C	0.02260			

Entropy Rates of a Stochastic Process

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- Example

Equally likely

$$H_{\max} = \log 27 \approx 4.755 \text{ bits/symbol}$$

independent

$$H_1(X) = -0.1817\log_2 0.1817 - 0.1073\log_2 0.1073 - \ldots - 0.00063\log_2 0.00063 \approx 4 \text{ bits/symbol}$$

Not independent

Combinations: th, he, in, er, an, re, ed, on ,es, st, en, at, to, nt, ha, nd, ou, ea, ng, as.

$$H_2(X) \approx 3.3 \text{ bits/symbol}$$

AEP for stationary stochastic processes

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- The significance of the entropy rate of a stochastic process arises from the AEP for a stationary ergodic process.
- $\{X_i\}$ a stationary ergodic process

$$-\frac{1}{n}\log P(X_1, X_2 \cdots X_n) \to H(\{X_i\}), \text{ with probability } 1$$

All the things (properties) we saw for AEP of i.i.d. processes hold, substituting $H(X)$ with $H(\{X_i\})$

when $\varepsilon \to 0, n \to \infty$

1) $\left|A_\varepsilon^{(n)}\right| \simeq 2^{nH(\{X_i\})}$ 2) $P\left(A_\varepsilon^{(n)}\right) \to 1$

3) $P(x_1, x_2 \cdots x_n) \simeq 2^{-nH(\{X_i\})}, \text{ if } (x_1, x_2 \cdots x_n) \in A_\varepsilon^{(n)}$

We can therefore represent the typical sequences of length n using approximately $nH(\{X_i\})$ bits, which shows the significance of the entropy rate as the average description length for a stationary ergodic process.

Summary

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- Asymptotic Equipartition Property (AEP)

AEP. "Almost all events are almost equally surprising." Specifically, if $X_1, X_2, \ldots$ are i.i.d. $\sim p(x)$, then

$$-\frac{1}{n}\log p(X_1, X_2, \ldots, X_n) \to H(X) \text{ in probability.} \qquad (3.28)$$

- Properties

1. If $(x_1, x_2, \ldots, x_n) \in A_\epsilon^{(n)}$, then $p(x_1, x_2, \ldots, x_n) = 2^{-n(H\pm\epsilon)}$.
2. $\Pr\left\{A_\epsilon^{(n)}\right\} > 1 - \epsilon$ for n sufficiently large.
3. $\left|A_\epsilon^{(n)}\right| \le 2^{n(H(X)+\epsilon)}$, where $|A|$ denotes the number of elements in set A.

Summary

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- Entropy Rates of a Stochastic Process

Entropy rate. Two definitions of entropy rate for a stochastic process are

$$H(\mathcal{X}) = \lim_{n \to \infty} \frac{1}{n} H(X_1, X_2, \ldots, X_n), \tag{4.76}$$

$$H'(\mathcal{X}) = \lim_{n \to \infty} H(X_n | X_{n-1}, X_{n-2}, \ldots, X_1). \tag{4.77}$$

For a stationary stochastic process,

$$H(\mathcal{X}) = H'(\mathcal{X}). \tag{4.78}$$

- AEP for Stationary Stochastic Processes